

Cassy Cormier

Professor Ayyagari

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1. What I Learned

Throughout this project, I gained hands-on experience building an AI-driven IIoT system from the ground up. Even though my original proposal focused on using fitness app data to warn users about carcinogenic foods, the Practical Path required me to shift toward a fully functional **Healthcare Sensor Dashboard** built in ThingsBoard Cloud. This shift taught me how to adapt my ideas to real technical constraints while still demonstrating AI, security, and autonomous decision-making.

I learned how telemetry flows through an IoT pipeline — from a simulated device, through secure authentication, into a cloud platform, and finally into a real-time dashboard. Creating a device in ThingsBoard and defining telemetry keys based on my dataset helped me understand how structured data becomes live sensor streams.

I also strengthened my understanding of **AI and autonomy in edge systems**. Even though I didn't train a full ML model, I implemented autonomous logic through thresholds, latency monitoring, and real-time insights. This showed me how AI doesn't always require a heavy model — sometimes simple rule-based intelligence is enough to demonstrate autonomous behavior.

On the development side, I learned how to work with tools like **VS Code, GitHub, FastAPI, Vue 3 + Vite, and ThingsBoard Cloud**, and how each tool fits into a modern IoT/AI workflow. I also deepened my understanding of dataset handling by downloading and preparing a Kaggle dataset, mapping its columns to telemetry keys, and integrating them into my dashboard.

Overall, this project helped me connect concepts from my AI coursework — data pipelines, cloud platforms, autonomy, and visualization — into one cohesive system.

2. Challenges I Faced

This project came with several technical challenges that pushed me to troubleshoot, iterate, and refine my approach.

One major challenge was **defining telemetry keys** correctly. The dataset contained many columns, and I had to translate them into ThingsBoard-compatible keys while ensuring the data types matched. If a key was misconfigured, the widget wouldn't update, which forced me to carefully validate each mapping.

Another challenge was **dashboard design**. ThingsBoard widgets can be sensitive to formatting, units, and time windows. I had to experiment with different chart types, gauges, and layouts to create a dashboard that was both readable and meaningful. This process taught me how to think like a UI/UX designer while still working within the constraints of an IoT platform.

Security was another important area. Configuring device credentials, understanding access tokens, and ensuring secure communication helped me appreciate how critical authentication is in IoT systems. Even in a simulated environment, I had to treat the device as if it were real hardware.

Finally, I faced the challenge of demonstrating **autonomous decision-making** without a full ML model. I solved this by implementing threshold-based logic and latency monitoring, which allowed the system to react to incoming data in real time. This experience helped me understand how autonomy can be layered into a system even before advanced AI models are added.

3. Insights & Future Directions

If I were to continue or expand this project, there are several improvements I would make.

First, I would integrate a **real predictive model** — for example, using Edge Impulse or a lightweight neural network — to analyze patterns in the fitness data and generate personalized recommendations. This would elevate the system from rule-based autonomy to true AI-driven intelligence.

Second, I would refine the dashboard by adding alerts, notifications, and more advanced visualizations. I would also explore ThingsBoard rule chains to automate actions such as sending warnings or adjusting thresholds dynamically.

Third, I would revisit my original idea of detecting carcinogenic food patterns. With more time, I could combine nutrition datasets with fitness telemetry to create a more holistic health-monitoring system.

Finally, I would focus on **real hardware integration**. Using a smartwatch, ESP32, or Raspberry Pi would allow me to test real sensor latency, battery constraints, and connectivity challenges — all of which are essential in real-world IoT deployments.

This project taught me how to design, build, and demonstrate a functioning AI-assisted IIoT system. I learned how to troubleshoot, adapt, and think critically about data, security, and autonomy. Most importantly, I gained confidence in my ability to take an idea from concept to implementation, even when the tools or requirements changed along the way.

